

**KOLEJ PROFESIONAL MARA BERANANG****DIPLOMA IN COMPUTER SCIENCE**

FINAL PROJECT REPORT

HOTEL MAINTENANCE MANAGEMENT SYSTEM (HMMS)

COURSE NAME: PROJECT | COURSE CODE: CSC2854

ACADEMIC SESSION: SESSION 1 2026/2027

Document Information	Details
Course Name	Project
Course Code	CSC2854
Academic Session	Session 1 2026/2027
Name	
I/D Number / Matric No.	
Class	DCS 6__
System	Hotel Maintenance Management System
Type of Assessment	Final Project
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I hereby declare that plagiarism in any form will not be tolerated in this assessment. Signature: _____

Date: _____

Assessment Period: 5 June 2026 - 3 July 2026

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1. Executive Summary

Testing was completed for the Hotel Maintenance Management System, a Laravel-based web application that supports authentication, dashboard monitoring, maintenance request management, task assignment, task status progression, notifications and profile management. The assessment combined manual functional validation, visual verification of the running application, source-level traceability and the existing automated regression suite.

Measure	Result
Documented functional test cases	18
Functional test cases passed	18
Functional test cases failed	0
Functional test cases blocked/not executed	0
Functional coverage	100%
Automated regression tests	25 passed (62 assertions)
Open critical or major defects	0
Overall readiness	READY FOR DEMONSTRATION

All documented test cases passed in the local test environment. No critical or major defect remained open at the reporting date. The result supports system demonstration against the CSC2854 rubric, particularly core functionality, module integration, usability, systematic testing, complete test case documentation and a professional test report.

2. Test Scope and Objectives

2.1 Objectives

- Verify that approved functional workflows operate accurately and consistently.
- Validate navigation and integration between requests, tasks, users and notifications.
- Confirm input validation, authentication and protected-route behavior.
- Document repeatable test cases with expected and actual outcomes.
- Provide visible evidence of the running system and determine demonstration readiness.

2.2 In Scope

Module	Functions Validated
Authentication	Valid/invalid login, logout, forgot password and protected routes
Dashboard	Request totals, status cards, recent request list and staff tasks
Maintenance Requests	Create, validate, list, view, edit and delete
Task Assignment	Assign staff, priority and notes; synchronize request status
Task Workflow	Start and complete task with timestamps and request status updates
Notifications	Generation, listing, individual read and mark-all-read
Profile	Profile update and password-related account functions
Technical Validation	Laravel routes, MySQL persistence and PHPUnit regression suite

2.3 Out of Scope

Production hosting, live email delivery, external service integration, high-volume load testing, penetration testing and cross-device hardware testing were not part of this local acceptance cycle.

3. Test Approach and Validation

A risk-based black-box approach was used for business workflows, supported by white-box inspection of routes, controllers, models and database migrations. Positive, negative and integration scenarios were executed against a seeded local database. Visual evidence was captured from the live application. Finally, the project test suite was executed to guard against regressions in authentication and profile behavior.

Technique	Purpose	Evidence
Positive functional testing	Confirm intended workflows succeed	TC-001, TC-003, TC-004, TC-006-TC-014
Negative testing	Confirm invalid or missing input is rejected	TC-002, TC-005
Security/navigation testing	Confirm authentication boundaries and logout	TC-015
Regression testing	Confirm existing application behavior remains stable	TC-018: 25 tests, 62 assertions
Visual validation	Confirm usability, status badges, forms and navigation	Figures A1-A5

Traceability review	Map tests to routes/controllers/database behavior	Section 6 and workbook register
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4. Test Environment

Environment Item	Configuration
Operating System	Microsoft Windows (local development workstation)
Application URL	http://127.0.0.1:8000
Browser	Chromium-based in-app browser, 1280 x 720 viewport
Application Server	Laravel development server on PHP
Framework	Laravel with Blade templates and Laravel Breeze authentication
Database	MySQL database: hotel_maintenance_db
Front End	Blade, Tailwind CSS/Vite assets and Font Awesome UI icons
Test Runner	PHPUnit through Laravel Artisan
Test Date	1 July 2026
Test Accounts	Administrator and maintenance staff accounts created for local validation
Test Data	Seeded requests in Pending, In Progress and Completed states

Test data was non-production data. Password values are intentionally excluded from this report. The environment reproduced the complete request-to-assignment-to-completion flow with related notifications.

5. Test Summary and Results

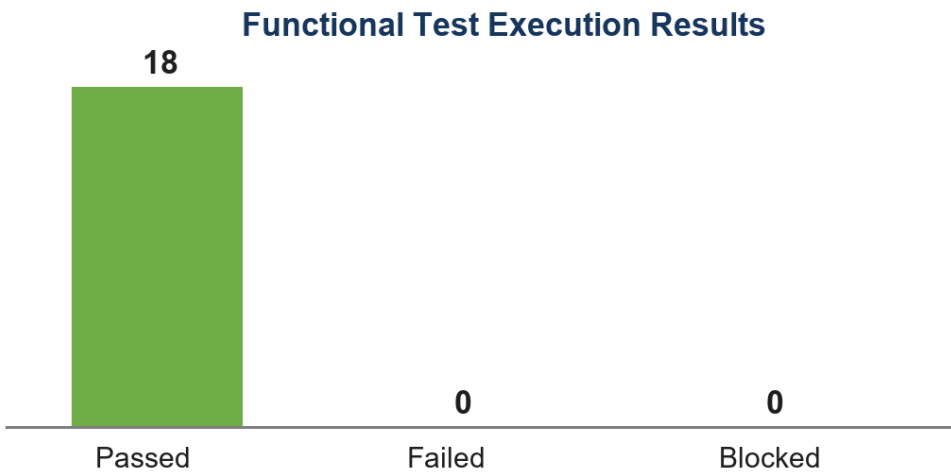


Figure 1. Functional test execution results

Status	Count	Percentage
Pass	18	100%
Fail	0	0%
Blocked / Not Executed	0	0%
Total	18	100%

The automated regression suite separately completed 25 tests with 62 assertions and zero failures. Functional cases and automated cases are reported separately to avoid double-counting coverage.

6. Requirements Traceability

Requirement Area	Implemented Component	Test Cases	Result
Secure user access	Authentication routes and middleware	TC-001, TC-002, TC-015, TC-016	Pass
Operational overview	DashboardController and dashboard view	TC-003	Pass
Request lifecycle	MaintenanceController and maintenance_requests table	TC-004-TC-008, TC-017	Pass
Staff assignment	MaintenanceTask model/controller and assignment routes	TC-009	Pass

Task progression	Start and complete task routes	TC-010, TC-011	Pass
User notification	NotificationController and notifications table	TC-012, TC-013	Pass
Account maintenance	Profile and password controllers	TC-014, TC-016	Pass
Technical stability	Laravel route set and PHPUnit suite	TC-018	Pass

Every in-scope requirement area is represented by at least one passing test case. Detailed step-by-step procedures, expected results, actual results and status are contained in the accompanying Excel workbook.

7. Defect Summary

Severity	Open	Resolved During Development	Status
Critical	0	0	None
Major	0	1	Resolved
Minor	0	2	Resolved
Total	0	3	No open defects

Defect ID	Issue	Root Cause	Resolution	Final Status
DF-001	Request/task relationship did not consistently use the request_id foreign key	Relationship mapping ambiguity	Explicit foreign keys were applied in the MaintenanceRequest and MaintenanceTask models	Closed
DF-002	Status badge styles could differ between underscore and hyphen forms	CSS class naming inconsistency	Both in_progress/in-progress and on_hold/on-hold variants are supported	Closed
DF-003	Duplicate role migration could fail on an existing role column	Schema evolution conflict	Migration checks Schema::hasColumn before adding or dropping the role column	Closed

No defect was found during final execution that prevented completion of an in-scope workflow. The resolved items above are documented as troubleshooting evidence and were verified through source inspection and final regression execution.

8. Error Detection and Troubleshooting

8.1 Troubleshooting Method

- Reproduce the problem with controlled local data.
- Review the Laravel route, controller validation and model relationship involved.
- Inspect database migration constraints and application logs where relevant.
- Apply the smallest targeted correction.
- Re-run the affected workflow and the full automated suite.
- Capture final visual evidence and record the outcome.

8.2 Example: Relationship Mapping

Symptom: task information could be missing or connected through the wrong default key. Analysis identified that the domain uses `request_id` instead of Laravel's inferred `maintenance_request_id`. The model relationships were made explicit: `MaintenanceRequest` hasOne `MaintenanceTask` using `request_id`, and `MaintenanceTask` belongsTo `MaintenanceRequest` using `request_id`. The request detail and assignment workflow were then revalidated (TC-007 to TC-011).

8.3 Example: Schema Compatibility

Symptom: a migration that adds the `role` column can conflict when the base `users` table already contains that column. The migration now guards the operation with `Schema::hasColumn`. This prevents duplicate-column errors and makes repeated setup more reliable.

8.4 Final Verification

The final regression execution reported 25 passed tests and 62 assertions. Live navigation produced successful pages for the dashboard, maintenance list, request detail, request creation form and notification centre. No blocking server or browser error was observed in the final evidence pass.

9. Conclusion and Readiness

The testing objectives were met. All 18 documented functional test cases passed, the automated suite passed 25 tests with 62 assertions, all in-scope requirements have traceable coverage, and no critical or major defect remains open. The system correctly supports the main hotel maintenance workflow from request submission through assignment, work progression, completion and notification.

Decision Area	Assessment
Functional correctness	Pass
Module integration and navigation	Pass
User interface and usability	Pass
Validation and protected access	Pass
Test documentation completeness	Pass
System readiness	READY FOR FINAL PROJECT DEMONSTRATION

Recommendation: proceed with the CSC2854 final project demonstration using the seeded administrator and staff test accounts. Retain the accompanying workbook as the primary step-by-step test case record.

Appendix A. System Evidence Screenshots

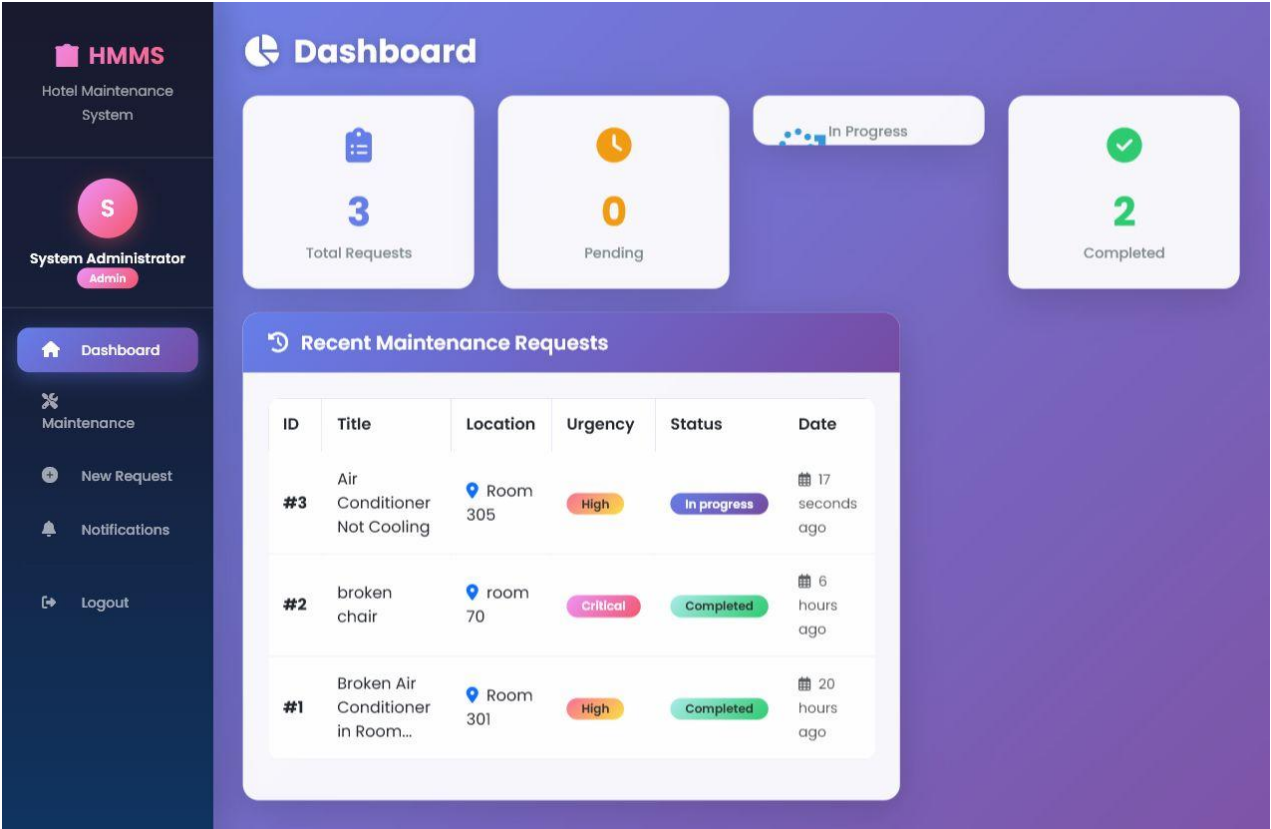


Figure A1. Administrator dashboard with request statistics and recent requests



Figure A2. Maintenance request register with urgency and status information

HMMS
Hotel Maintenance System

S
System Administrator
Admin

Dashboard

Maintenance

New Request

Notifications

Logout

Request Details

#3 In progress High

Air Conditioner Not Cooling

Location: Room 305

Description:

The air conditioner runs but does not cool the room.

Reported By: System Administrator

Date: 01 Jul 2026 15:08

Task Details

Assigned To: Maintenance Staff **Task Status:** Pending

Priority: High **Notes:** Inspect filter, thermostat, and refrigerant level.

Edit

Figure A3. Request detail and task assignment information

HMMS
Hotel Maintenance System

S
System Administrator
Admin

Dashboard

Maintenance

New Request

Notifications

Logout

+ New Maintenance Request

H Title

e.g., Broken Air Conditioner

Description

Describe the issue in detail...

Location

e.g., Room 301, Lobby, Elevator B

Urgency Level

Medium

Image (Optional)

Choose File No file chosen

Figure A4. New maintenance request form

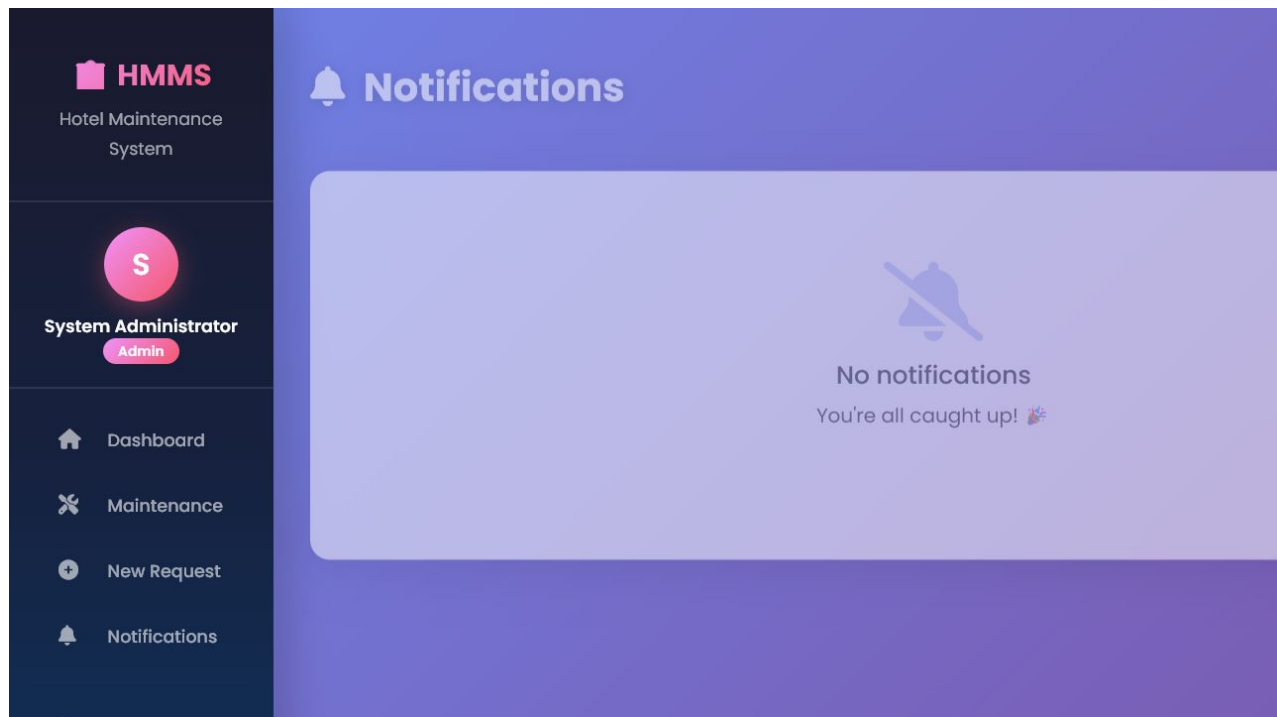


Figure A5. Notification centre

Appendix B. Test Case Register

The complete test case register is supplied as HMMS_FULL_TEST_CASES.xlsx. It contains a summary worksheet and 18 individual test case worksheets. Every worksheet records prerequisites, test data, three execution steps, expected results, actual results and final status.

Workbook Section	Content
TEST SUMMARY	18 cases, module mapping, status and evidence reference
TC-001 to TC-002	Authentication positive and negative scenarios
TC-003	Dashboard verification
TC-004 to TC-008	Maintenance request lifecycle
TC-009 to TC-011	Assignment and task progression
TC-012 to TC-013	Notification behavior
TC-014	Profile update
TC-015 to TC-016	Protected navigation and password reset
TC-017	Request deletion
TC-018	Automated regression execution

Appendix C. CSC2854 Assessment Brief and Compliance Mapping

Source document: FYP CSC2854 S1 2026 2027. The following section consolidates the assessment requirements supplied with the project and maps them to the evidence contained in this combined report.

C.1 Assessment Information

Item	Requirement
Institution	Kolej Profesional MARA Beranang
Programme	Diploma in Computer Science
Course	CSC2854 - Project
Academic Session	Session 1, 2026/2027
Assessment	Final Project
Assessment Period	5 June 2026 to 3 July 2026
Course Learning Outcome	CLO3: Build a functional system based on user and technical requirements.
Report Format	Arial, 11 pt, 1.5 line spacing
Required Supporting Material	Source code and a 2-3 minute error detection and troubleshooting video

C.2 System Implementation Requirements

Rubric Area	Required Demonstration	HMMS Evidence
1(a) Requirement Fulfilment and Scope	Show that approved user and technical requirements are implemented.	Sections 2 and 6; Appendix B; TC-001 to TC-018
1(b) Construction and Core Functionality	Execute major functions accurately and consistently without critical errors.	Sections 3 and 5; Figures A1-A5; full test cases
1(c) Integration and Navigation	Show seamless workflows and module interaction.	Request, assignment, task and notification traceability in Section 6
1(d) User Interface and Usability	Demonstrate a clear, consistent and usable interface.	Appendix A system screenshots
1(e) Technical Implementation	Describe frameworks, languages, database and development tools.	Section 4 test environment and technical validation

C.3 System Testing Requirements

Rubric Area	Required Demonstration	HMMS Evidence
2(a) Testing and Validation	Explain systematic testing and prove validation against requirements.	Sections 3, 5 and 6

2(b) Error Detection and Troubleshooting	Identify, analyse, resolve and verify system errors.	Sections 7 and 8; controlled relationship bug demonstration
2(c) Test Case Documentation	Document detailed tests covering major functions and outcomes.	Appendix D contains all 18 detailed test cases
2(d) Test Report	Provide summary, environment, defect summary, conclusion and suitable charts.	Sections 1, 4, 5, 7 and 9

C.4 Submission Checklist

- Working and functional Hotel Maintenance Management System.
- Appropriate digital technologies: Laravel, PHP, Blade, MySQL and PHPUnit.
- Practical development evidence through integrated request and task workflows.
- System testing evidence through 18 documented functional cases and an automated regression suite.
- Supporting technical documentation consolidated into this report.
- Error detection and troubleshooting video supported by the prepared controlled bug scenario.

Appendix D. Complete Test Case Documentation

This appendix reproduces the complete content of the accompanying Excel test case workbook so that the assessment brief, test report, screenshots and detailed test cases are available in one Word document.

TC-001: Valid administrator login

Field	Details
Test Case ID	TC-001
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Valid administrator login
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	admin@hotel.test / valid password

Step	Action	Expected Result	Actual Result	Status
1	Open the login page	Login page is displayed	As expected	Pass
2	Enter valid administrator credentials	Credentials are accepted	As expected	Pass
3	Select Log in	Dashboard is displayed with the administrator role	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-002: Invalid login is rejected

Field	Details
Test Case ID	TC-002
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Invalid login is rejected
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	valid email / invalid password

Step	Action	Expected Result	Actual Result	Status
1	Open the login page	Login page is displayed	As expected	Pass
2	Enter a valid email and invalid password	Credentials can be entered	As expected	Pass
3	Select Log in	Validation message is shown and access is denied	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-003: Dashboard statistics and recent requests

Field	Details
Test Case ID	TC-003
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Dashboard statistics and recent requests
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Seeded requests in multiple states

Step	Action	Expected Result	Actual Result	Status
1	Log in as administrator	Dashboard loads	As expected	Pass
2	Open Dashboard	Total, Pending, In Progress and Completed cards appear	As expected	Pass
3	Compare summary cards with recent request rows	Counts and recent request statuses are consistent	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-004: Create maintenance request

Field	Details
Test Case ID	TC-004
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Create maintenance request
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Title, description, Room 305, High

Step	Action	Expected Result	Actual Result	Status
1	Open New Request	Request form is displayed	As expected	Pass
2	Enter all required request details	Valid data is accepted	As expected	Pass
3	Submit the form	Request is saved as Pending and success feedback appears	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-005: Required-field validation

Field	Details
Test Case ID	TC-005
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Required-field validation
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Blank required fields

Step	Action	Expected Result	Actual Result	Status
1	Open New Request	Request form is displayed	As expected	Pass
2	Leave required fields blank	Blank data remains unsubmitted	As expected	Pass
3	Submit the form	Validation feedback appears and no request is created	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-006: View maintenance request list

Field	Details
Test Case ID	TC-006
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	View maintenance request list
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Existing maintenance records

Step	Action	Expected Result	Actual Result	Status
1	Open Maintenance	Paginated list is displayed	As expected	Pass
2	Review table columns and status badges	ID, title, location, urgency, status and actions are visible	As expected	Pass
3	Open a listed request	Selected request detail page opens	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-007: View maintenance request detail

Field	Details
Test Case ID	TC-007
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	View maintenance request detail
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Request #3

Step	Action	Expected Result	Actual Result	Status
1	Open request #3	Detail page loads	As expected	Pass
2	Review request information	Requester, description, location, urgency and status are correct	As expected	Pass
3	Review task assignment panel	Assignment information and available actions are visible	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-008: Edit maintenance request status

Field	Details
Test Case ID	TC-008
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Edit maintenance request status
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Request #3 / On Hold

Step	Action	Expected Result	Actual Result	Status
1	Open request #3 edit page	Edit form is displayed	As expected	Pass
2	Change status to On Hold	Allowed status is accepted	As expected	Pass
3	Save changes	Updated status is persisted and shown in the list	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-009: Assign request to staff

Field	Details
Test Case ID	TC-009
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Assign request to staff
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	staff@hotel.test / High priority

Step	Action	Expected Result	Actual Result	Status
1	Open an unassigned request	Assignment controls are available	As expected	Pass
2	Choose Maintenance Staff and High priority	Valid staff and priority are accepted	As expected	Pass
3	Submit assignment	Task is created and request becomes In Progress	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-010: Start assigned task

Field	Details
Test Case ID	TC-010
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Start assigned task
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Pending assigned task

Step	Action	Expected Result	Actual Result	Status
1	Log in as assigned staff	Assigned task is visible	As expected	Pass
2	Open assigned task	Task detail is displayed	As expected	Pass
3	Select Start Task	Task and request become In Progress with started timestamp	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-011: Complete assigned task

Field	Details
Test Case ID	TC-011
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Complete assigned task
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	In-progress task

Step	Action	Expected Result	Actual Result	Status
1	Open an in-progress assigned task	Task can be opened	As expected	Pass
2	Select Complete Task	Completion action succeeds	As expected	Pass
3	Return to dashboard	Task and request show Completed with completion timestamp	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-012: Notification generated on assignment

Field	Details
Test Case ID	TC-012
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Notification generated on assignment
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Assignment for request #3

Step	Action	Expected Result	Actual Result	Status
1	Assign request to staff	Assignment completes	As expected	Pass
2	Log in as assigned staff	Unread indicator is available	As expected	Pass
3	Open Notifications	Assignment message links to the correct request	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-013: Mark notifications as read

Field	Details
Test Case ID	TC-013
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Mark notifications as read
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Unread staff notification

Step	Action	Expected Result	Actual Result	Status
1	Open Notifications	Notifications list loads	As expected	Pass
2	Mark one notification as read	Selected item becomes read	As expected	Pass
3	Use Mark All as Read when applicable	Unread count becomes zero	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-014: Profile information update

Field	Details
Test Case ID	TC-014
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Profile information update
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Updated display name

Step	Action	Expected Result	Actual Result	Status
1	Open Profile	Profile form loads	As expected	Pass
2	Change the display name	Valid name is accepted	As expected	Pass
3	Save profile	Success feedback appears and new name persists	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-015: Application route protection

Field	Details
Test Case ID	TC-015
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Application route protection
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Unauthenticated browser session

Step	Action	Expected Result	Actual Result	Status
1	Log out	Session is ended	As expected	Pass
2	Navigate directly to /dashboard	Dashboard redirects to Login	As expected	Pass
3	Navigate directly to /maintenance	Maintenance redirects to Login	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-016: Password reset request validation

Field	Details
Test Case ID	TC-016
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Password reset request validation
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Registered user email

Step	Action	Expected Result	Actual Result	Status
1	Open Forgot Password	Reset form loads	As expected	Pass
2	Enter a registered email address	Email is accepted	As expected	Pass
3	Submit reset request	System returns a reset-link status without exposing account data	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-017: Delete maintenance request

Field	Details
Test Case ID	TC-017
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Delete maintenance request
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	Disposable test request

Step	Action	Expected Result	Actual Result	Status
1	Create a disposable request	Disposable request is created	As expected	Pass
2	Select Delete and confirm	Delete action succeeds	As expected	Pass
3	Search the request list	Deleted request no longer appears	As expected	Pass

Result: PASS - All expected outcomes were achieved.

TC-018: Automated regression suite

Field	Details
Test Case ID	TC-018
Version	1.0
Date Tested	01 July 2026
Final Status	Pass
Test Scenario	Automated regression suite
Prerequisite 1	Application running at http://127.0.0.1:8000
Prerequisite 2	Supported browser and appropriate test account
Test Data	PHPUnit feature and unit tests

Step	Action	Expected Result	Actual Result	Status
1	Run the project test suite	Test runner starts without configuration error	As expected	Pass
2	Review all test groups	Authentication, profile and application tests complete	As expected	Pass
3	Confirm assertion summary	25 tests and 62 assertions pass with zero failures	As expected	Pass

Result: PASS - All expected outcomes were achieved.